Quiz 6

Friday, October 16 — matrices, Ax, and transformations (Week 7)

Last updated: October 5, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Covers the week before only: Matrices, Ax, Transforms, and Lab 3 (Week 7). Same-day RREF is **not** on this quiz.

Textbooks: Boyd & Vandenberghe (VMLS) · Lachniet Octave PDF · also on Resources.

Schedule

Core

Fluency (complete prompts — practice these by hand)

C6.1. Let

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 1 \end{bmatrix}.$$

(a) What are the dimensions of A? Is it tall, wide, or square? (b) What is A_{23}? Write the second column as a vector.

C6.2. For each matrix, list every label that applies from: zero, identity, diagonal, upper triangular, lower triangular, symmetric, sparse (tiny examples: call it sparse if at least half the entries are zero).

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

C6.3. Compute $A + A^{\mathsf{T}}$ for $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$. Is the result symmetric? Why or why not (one sentence)?

C6.4. Compute

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

two ways: (i) row-inner-product view; (ii) as a linear combination of the columns. Confirm the answers match.

C6.5. Let A have columns $a_1 = (1, 0, 1)$ and $a_2 = (0, 1, 1)$. Write A explicitly. For $x = (3, -2)$, write Ax as a linear combination of a_1, a_2, then evaluate.

C6.6. True or false, with a one-line reason: if A is $m \times n$, then Ax always lies in the span of the columns of A.

C6.7. A sensor maps a 3-vector of raw readings x to a 2-vector of summaries via $y = Ax$. What are the dimensions of A? In one sentence, what does the first column of A mean?

C6.8. Reflection through the x_1-axis sends $e_1 = (1, 0) \mapsto (1, 0)$ and $e_2 = (0, 1) \mapsto (0, -1)$. Write the matrix A and compute $A(2, 3)$.

C6.9. Write the 2×2 matrix for rotation by 90° counterclockwise. Where do $(1, 0)$ and $(0, 1)$ go?

C6.10. Let $X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (columns are three points) and $A = \mathrm{diag}(2, 1)$. Compute AX and describe the geometric effect in one sentence.

C6.11 (Lab 3). Predict where $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ sends the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$. Then check with a short Octave or Python computation (or a careful sketch).

Textbook Core

- B 6.1, 6.2, 6.4, 6.6
- B 7.2

Stretch

- B 6.7, 6.8, 6.9, 6.10 (longer modeling / interpretation)
- B 6.12, 7.1, 7.3, 7.4, 7.8

What a quiz item might change

A small matrix or two points, which geometric map is used, whether you must name a matrix type, or whether the prompt asks for the column interpretation of Ax.